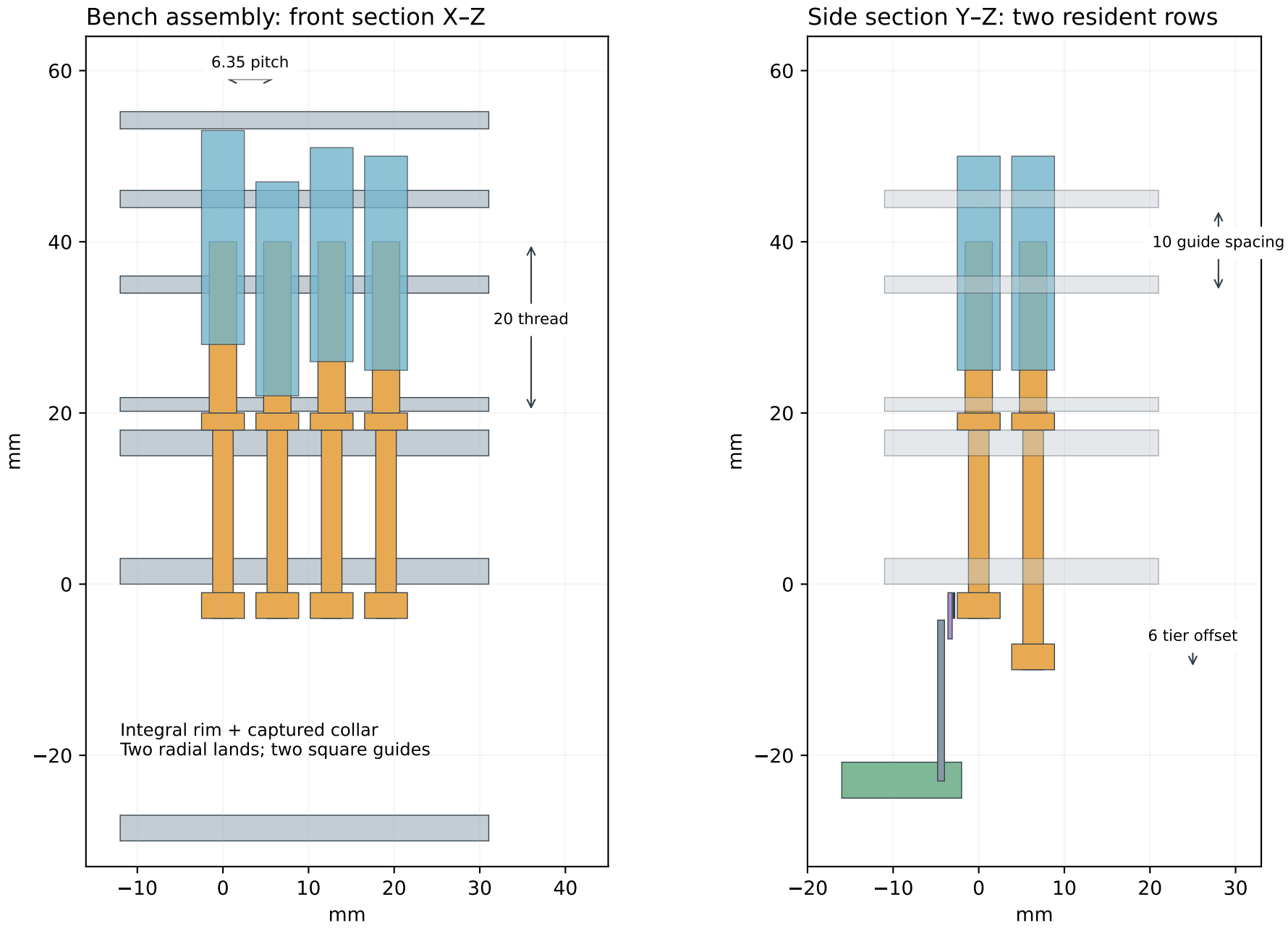
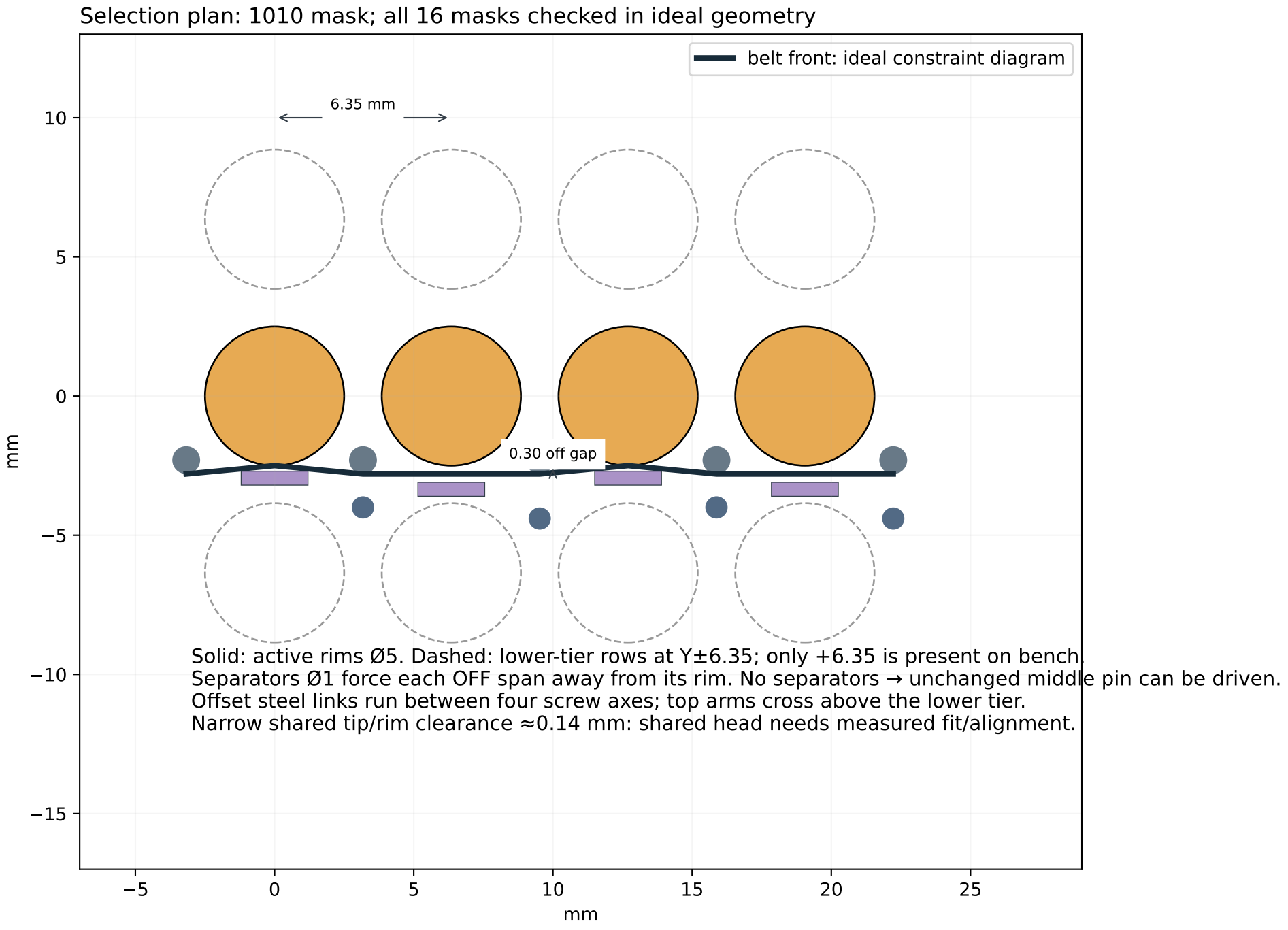
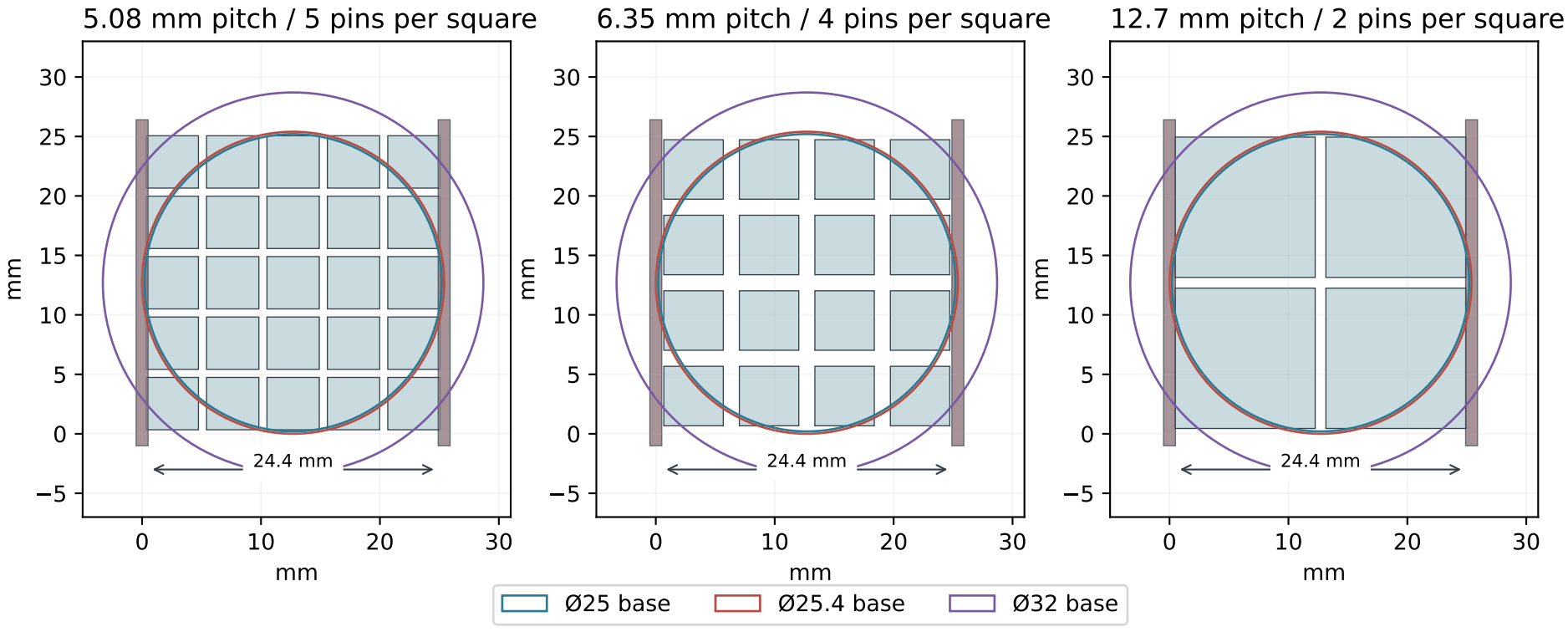
DES-007 • mm • nominal geometry, NOT a qualified machine



Prototype usable q = 0…6 mm; nut bottom = 22+q; top = 47+q.
Roof stops upward travel at q≈6.2. Downward collar stop is q≈−2. Never drive either stop.

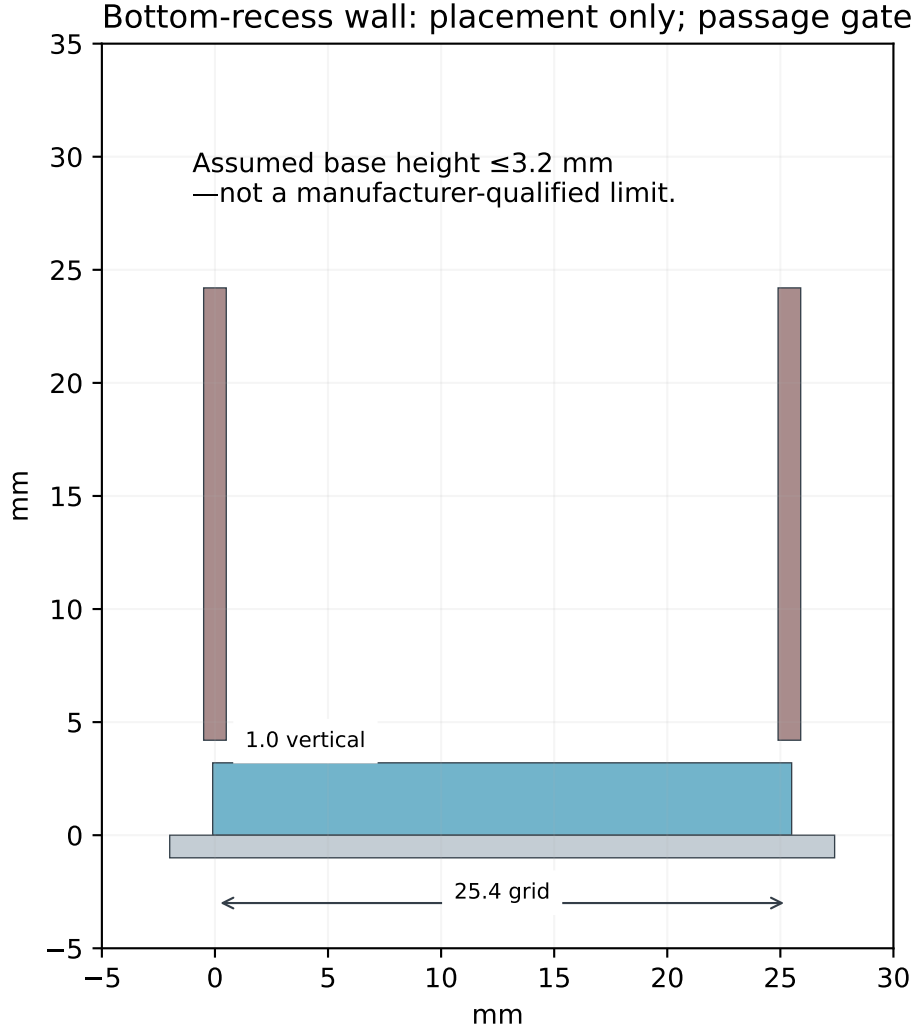

Selection plan: 1010 mask; all 16 masks checked in ideal geometry
belt front: ideal constraint diagram
6.35 mm
0.30 off gap
Solid: active rims Ø5. Dashed: lower-tier rows at Y±6.35; only +6.35 is present on bench.
Separators Ø1 force each OFF span away from its rim. No separators → unchanged middle pin can be driven.
Offset steel links run between four screw axes; top arms cross above the lower tier.
Narrow shared tip/rim clearance ≈0.14 mm: shared head needs measured fit/alignment.
mm
mm
10
5
0
−5
−10
−15
−5
0
5
10
15
20
25

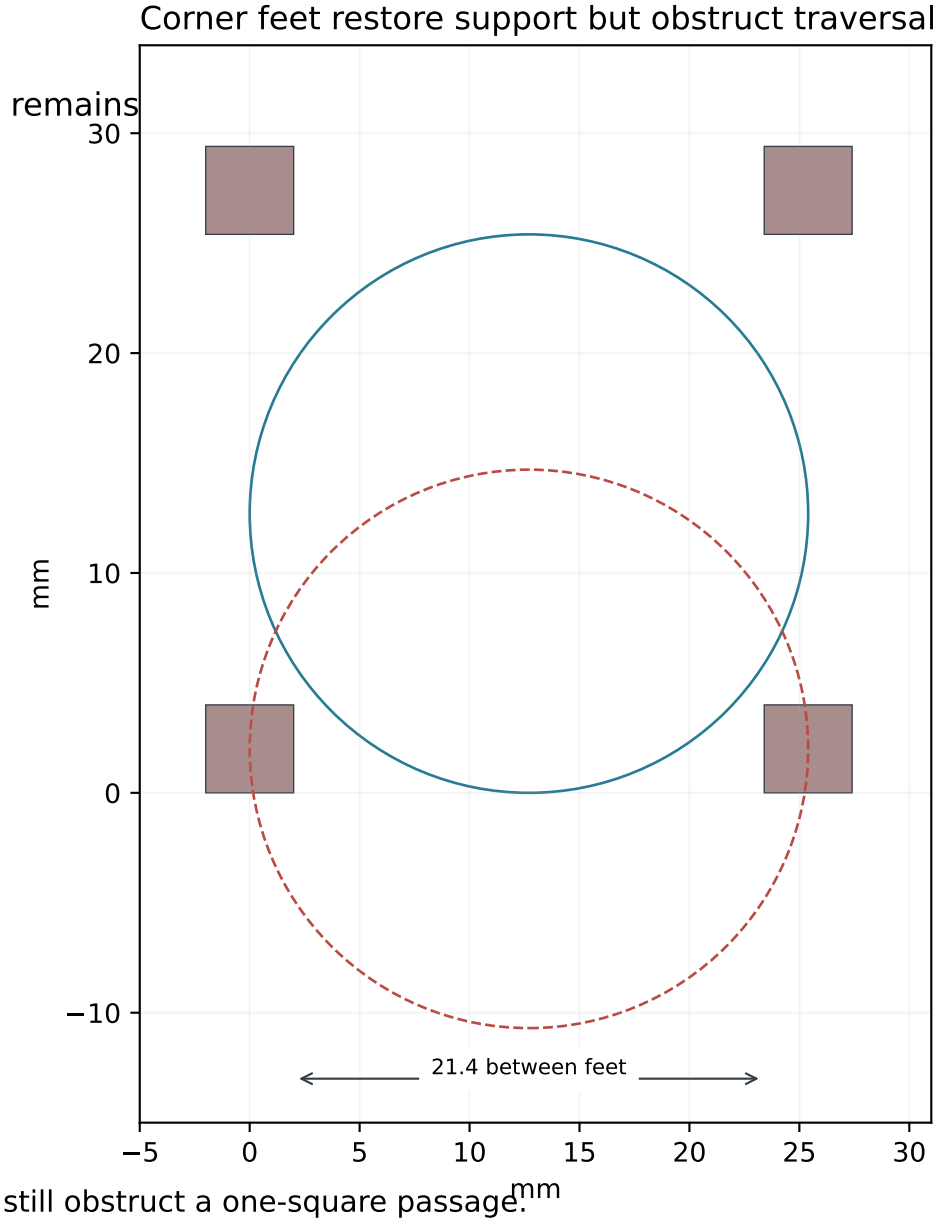
A finer pin pitch does not make grid-line walls zero thickness



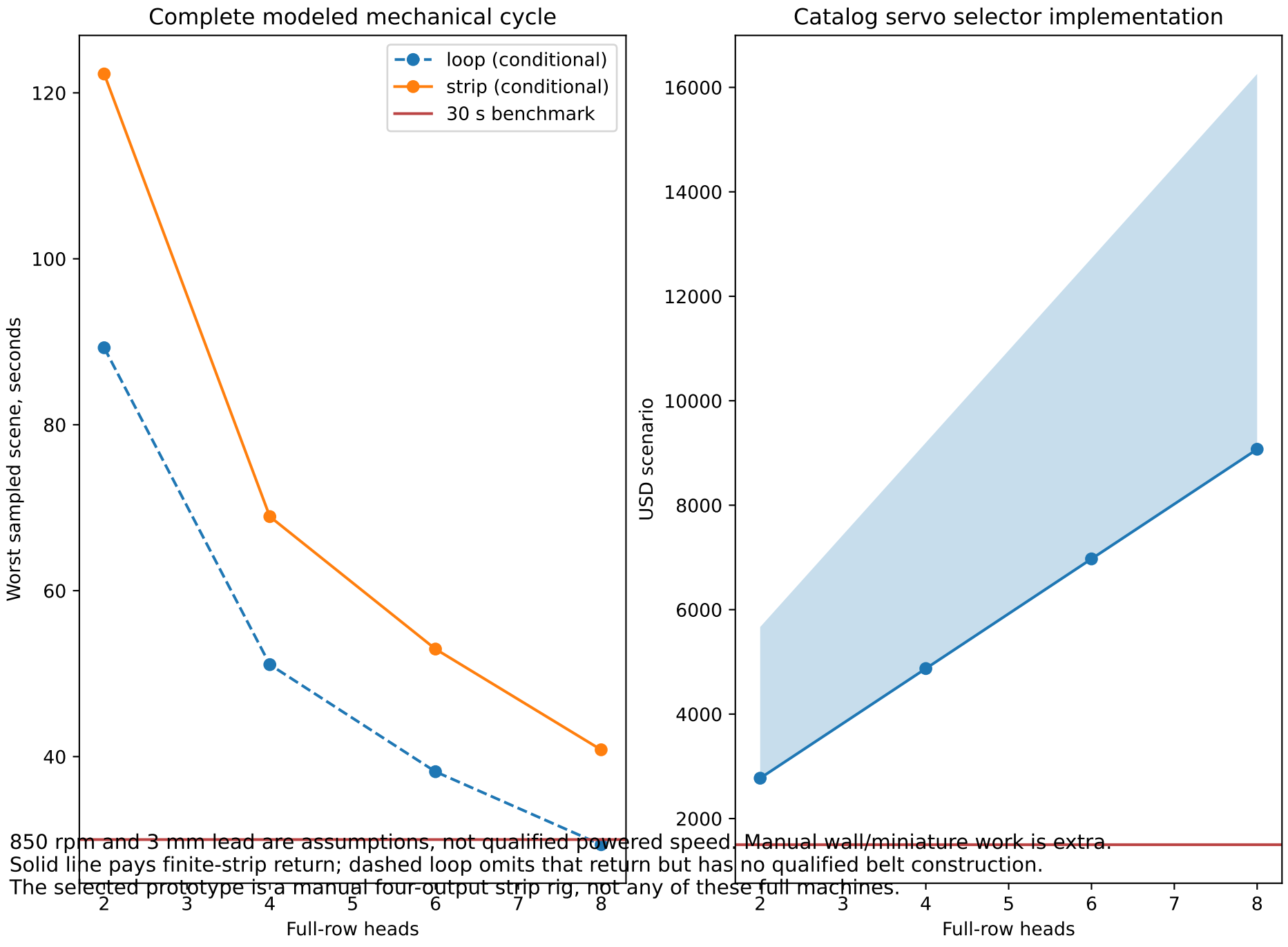
25 mm bases are manufacturer-specified by WizKids. One-inch bases also exist.
At 1 mm centered wall thickness: 24.4 mm nominal, 24.0 mm with 0.2 mm inward error per wall.
Allowing Ø25.6 mm base envelope + 0.25 mm placement per side needs ≥26.1 mm clear.
A 0.4 mm wall leaves 25.0 mm nominal: still no tolerance/placement margin for a 25 mm base.

Wall iteration: preserve the failed traversal witness





An undercut can admit a centered base, but supporting posts still obstruct a one-square passage.
Do not call this a solved wall system. A wider opening or displaced supports requires an explicit layout trade-off.
Doors need the same actual-clearance check. Oversized 32 mm bases also overlap neighboring one-inch squares.


More parallel heads do not establish an affordable solution
Complete modeled mechanical cycle
loop (conditional)
strip (conditional)
30 s benchmark
Worst sampled scene, seconds
120
100
80
60
40
2
3
4
5
6
7
8
Full-row heads
Catalog servo selector implementation
USD scenario
16000
14000
12000
10000
8000
6000
4000
2000
Full-row heads
850 rpm and 3 mm lead are assumptions, not qualified powered speed. Manual wall/miniature work is extra.
Solid line pays finite-strip return; dashed loop omits that return but has no qualified belt construction.
The selected prototype is a manual four-output strip rig, not any of these full machines.